Anthropogenic Evolution

A Proposed Third Driver of Evolutionary Change

Version 2

Tirth Kapadia | Independent Researcher | India | 2026

Abstract

Classical evolutionary theory identifies two primary drivers of biological change: natural selection, as described by Charles Darwin, and random genetic mutation. This paper proposes a third driver, termed Anthropogenic Evolution, defined as the directed modification of evolutionary trajectories through human cultural, technological, and ideological agency. Unlike natural selection, which operates through environmental pressure without intent, Anthropogenic Evolution introduces choice as a mechanism capable of overriding, redirecting, or accelerating the evolutionary process. Version 2 extends the original framework in three directions: a challenge to Gause's competitive exclusion principle and MacArthur's niche theory as applied to intraspecies competition; an analysis of the replacement of Darwinian reproductive fitness as the operative selection directive with manufactured status competition; and an examination of the 2020 inflection point as an acceleration event that made the mismatch between biological baseline and cultural environment visible at the level of population health.

1. Defining Anthropogenic Evolution

Anthropogenic Evolution is proposed as the directed modification of evolutionary trajectories through human agency operating at cultural, technological, and ideological levels. It is distinguished from natural selection by three properties:

Intent

Anthropogenic Evolution involves deliberate human decisions, whether individual or collective, that alter which traits persist in a population. The decision may be conscious or may emerge from cultural systems without individual awareness of its evolutionary consequences.

Speed

Natural selection operates across thousands to millions of generations. Anthropogenic Evolution can produce measurable population-level changes within decades. This acceleration places it in a different category from classical evolutionary mechanisms.

Reversibility

Natural selection is directional and largely irreversible on short timescales. Anthropogenic Evolution can be reversed, redirected, or halted through changes in human behaviour, culture, or technology.

2. The Classical Framework and Its Limitations

Darwin's theory of natural selection, published in 1859, established that organisms better adapted to their environment survive and reproduce at higher rates, passing favourable traits to subsequent generations. Mendel's genetics provided the mechanism, and the Modern Synthesis of the 20th century united both into a coherent framework: random mutation generates variation, natural selection filters it, and populations change over time.

Lamarck's earlier proposal that traits acquired through use or disuse during an organism's lifetime are inherited by offspring was correctly rejected as the primary mechanism of evolutionary change. However, Lamarck's intuition that what organisms do shapes what they become contains a truth his mechanism failed to capture. The error was not in the observation but in the level of description.

What Lamarck observed, and Darwin's framework does not fully account for, is the role of agency in directing evolutionary pressure. In all non-human species, the environment applies pressure without negotiation. No organism chooses its selective landscape. Humans are the exception. Humans build the environment that selects them.

2.1 The Parallel Problem

Convergent evolution describes separate lineages independently developing similar traits in response to similar environmental pressure. Divergent evolution describes a shared ancestral trait splitting into distinct forms as lineages separate. Both models assume the trait in question solves an external environmental problem. What the fossil record of human cognition shows does not fit neatly into either framework.

Cranial capacity expansion, intentional burial of the dead, ochre use indicating symbolic thought, and tool complexity all appear across multiple hominin species within the same geological window. These traits did not emerge in one lineage and spread outward. They appeared in parallel, across separate populations, at similar points in time.

The most parsimonious explanation is that consciousness and metacognition are threshold phenomena. Once neural complexity crosses a certain point, certain cognitive outputs emerge architecturally. What made Homo sapiens the survivor was not superior intelligence or larger brain volume alone. It was adaptability, the capacity to recalibrate behaviour rapidly in response to changing conditions.

3. Evidence and Case Studies

Dietary ideology and dental reduction

Human canine teeth evolved for tearing meat. In populations sustaining plant-based diets across generations, canine teeth face no functional demand. Selection pressure for their prominence weakens and the hypothesis predicts measurable reduction over sufficient generations. The driver in both cases is not environment and not random mutation. It is human dietary choice.

Lactose tolerance and intolerance

Lactose intolerance is not a deficiency. It is the normal mammalian state. Every mammal stops producing lactase after weaning. Lactose tolerance in adults is the mutation. It spread because humans made a choice: keep cattle. That single cultural decision reshaped the genetics of entire populations [Tishkoff et al., Nature Genetics, 2007]. 70% of the world population is lactose intolerant. The 30% who are tolerant carry a mutation that spread because their ancestors chose to drink milk.

Appendix reduction

The appendix is called vestigial. But it is vestigial only in populations that made it so. In populations with high plant-fibre diets, the appendix plays an active role in gut microbiome support. The reduction is not universal. It is dietary. It is anthropogenic.

Biotechnology as explicit anthropogenic evolution

CRISPR-Cas9 gene editing technology represents the most direct expression of Anthropogenic Evolution. For the first time in the history of life on Earth, an organism can deliberately and precisely alter its own genome and the genomes of its offspring. Natural selection is no longer the only mechanism by which heritable change occurs [Liang et al., Protein and Cell, 2015; He et al., 2018].

4. The Lamarck Rehabilitation

Lamarck was wrong about the mechanism. He was not wrong about the intuition. His proposal that what organisms do shapes what they become has been partially vindicated through epigenetics. Epigenetic research has demonstrated that environmental experiences can alter gene expression in ways that are heritable for one to three generations. Famine during pregnancy alters metabolic gene expression in grandchildren. Trauma alters DNA methylation patterns that can be transmitted to offspring [Jablonka and Lamb, 2005; Skinner et al., 2010].

Anthropogenic Evolution operates through multiple channels: Darwinian selection shaped by human-created environments, epigenetic inheritance shaped by human-created stressors, and direct genomic editing bypassing both.

5. Implications

If Anthropogenic Evolution is recognised as a distinct evolutionary driver, several implications follow. Evolutionary biology must account for cultural and technological systems as selective environments. The rate of human biological change is accelerating beyond anything the classical framework was designed to model. Ethical frameworks governing biotechnology are, in effect, governance of evolution itself. The distinction between the natural and the artificial becomes biologically meaningless.

Human dietary choice represents a selection environment of cultural origin. The evolutionary consequences of that choice are Darwinian in mechanism but anthropogenic in cause. Natural selection remains the filter. Anthropogenic Evolution determines what passes through it.

6. The Original Hybrid

Anthropogenic Evolution is typically located in the present. But its origin may be considerably older. Genomic sequencing has confirmed that modern humans carry between one and four percent Neanderthal DNA, with some populations carrying additional Denisovan contributions [Green et al., Science, 2010; Reich et al., Nature, 2010].

The first anthropogenic evolutionary event was not a technology. It was a meeting. The mechanism is Darwinian. The driver was not environmental pressure but interspecies encounter. The emergence of the metacognitive threshold was a discontinuous event: a genetic recombination that crossed a threshold no prior hominid had reached.

7. The Cost of Becoming the Selector

Natural selection is indifferent. It does not intend outcomes, it filters them. Anthropogenic Evolution introduced something natural selection never had: intent. And with intent came consequences that nature, left alone, would never have produced.

The most immediate is antibiotic resistance. Penicillin was discovered in 1928. By 1940, bacteria resistant to it had already been documented. Humans deployed antibiotics at industrial scale and created the strongest selection pressure bacteria had ever faced. The chronic disease epidemic tells a parallel story. Human biology evolved across millions of years of scarcity. Anthropogenic Evolution produced permanent abundance in under two centuries. The biology did not recalibrate. Type 2 diabetes, cardiovascular disease, and obesity are diseases of evolutionary mismatch.

The sixth mass extinction is perhaps the starkest consequence. Five previous mass extinctions were caused by asteroid impact, volcanic activity, or gradual atmospheric change. The sixth is being caused by a single species making deliberate choices about land use, resource extraction, and economic organisation.

The human body is among the most adaptable biological systems ever produced by natural selection. That adaptability is now working against the species. We adapted to sedentary behaviour. We adapted to artificial light disrupting circadian rhythms. We adapted to nutritional monotony. We adapted to chronic low-grade stress as a baseline state. Each adaptation solved a short-term problem and created a long-term one.

8. The Collapse of Competitive Ecology

Gause's competitive exclusion principle, formalised in 1934, holds that two species competing for the same limited resource cannot stably coexist. MacArthur's niche theory extended this framework by arguing that species differentiate their resource use to reduce direct competition. Both frameworks assume that competition is for physical resources: food, territory, mates, water.

Anthropogenic Evolution has produced a species for which these frameworks no longer adequately apply. For the majority of the global population in industrialised societies, physical resource scarcity is no longer the operative selective pressure. The competition that now shapes behaviour, drives hormonal stress responses, determines social hierarchies, and produces the chronic psychological strain documented across modern populations is not competition for survival resources. It is competition for status.

The niche that humans now compete for is not food or territory but recognition, achievement, and social position. This competition is manufactured. Cultural systems, educational institutions, economic structures, and media environments have collectively produced a selective landscape in which the organism's perceived survival depends on outperforming other members of the same species across metrics that have no direct relationship to biological fitness.

Unlike competition for physical resources, status competition has no natural limit. Food scarcity resolves when enough food exists. Status competition is structurally infinite because the comparative nature of status means that every achievement by one individual resets the threshold for others. The goalpost moves with the population. The competition cannot be won because the definition of winning changes continuously. This produces a chronic activation of the stress response in organisms whose physiology evolved to handle acute, resolvable threats, not sustained, irresolvable ones.

Gause and MacArthur described competition for resources in a world where resources were the constraint. Anthropogenic Evolution has made attention and status the constraint and produced a competitive landscape that neither framework was designed to model.

9. The Replacement of the Darwinian Directive

Darwin's selection criterion is reproductive fitness. The organism that survives and reproduces at higher rates passes its traits to subsequent generations. The body was built around this directive. Eat. Drink. Sleep. Reproduce. Everything else was instrumental to this directive even when it did not appear to be.

Anthropogenic Evolution has replaced the operative directive without replacing the biological hardware running it. The cultural environment now selects for achieve, accumulate, and stand out. These are not the same directive. And critically, many of the behaviours the manufactured directive produces actively reduce reproductive fitness by the Darwinian measure.

Delayed reproduction in populations pursuing educational and career advancement. Declining birth rates in the most economically developed societies, precisely where the achievement directive is most intensively installed. Chronic stress reducing fertility. Processed food compromising hormonal balance and reproductive capacity. The species is being selected, by its own cultural environment, against the directive that natural selection built it for.

Furthermore, the same species has begun manipulating its own members for competitive advantage at scale. Advertising systems exploit limbic responses to manufactured desire. Political systems exploit fear responses to manufactured threat. Social media algorithms exploit social bonding drives to generate engagement that serves the algorithm rather than the organism. This is intraspecies manipulation operating through the same neurobiological channels that evolved for genuine survival information. No other species does this at scale.

10. The 2020 Inflection Point

The processes described in this paper have been underway for centuries. What 2020 represents is an acceleration event that made the mismatch between biological baseline and cultural environment visible at the level of population health.

The COVID-19 pandemic and its aftermath produced several conditions simultaneously: enforced disruption of normal social structures, mass transition to digital environments, sustained uncertainty generating chronic stress activation, pharmaceutical interventions at unprecedented population scale, and the elimination of physical movement patterns the body depends on for metabolic regulation.

What is notable from the Anthropogenic Evolution framework is the acceleration of gene expression dysregulation in younger populations. Type 2 diabetes, previously a disease of middle and old age, is now documented in teenagers. Hormonal imbalances that affect reproductive capacity and psychological stability are presenting at ages where they were previously rare. Inflammatory conditions associated with chronic stress are appearing earlier in the lifespan.

Evolution operates on generational timescales. The mismatch documented here is operating on a timescale of years and decades. Natural selection cannot correct the problem because natural selection cannot work fast enough. The species has produced conditions that are changing its biology faster than biology can adapt. Anthropogenic Evolution has outpaced the mechanism it was supposed to work within. That is not a theoretical concern. It is a current, measurable, population-level event.

11. Conclusion

Darwin described how life changes without agency. Lamarck intuited that agency matters but described the mechanism incorrectly. Anthropogenic Evolution proposes that both were partially right and that the full picture requires a third category: directed evolutionary change through human choice operating at cultural, ideological, and technological scales.

Version 2 extends this framework to account for three developments the original paper did not address. First, that classical competitive ecology frameworks fail to model intraspecies status competition, which now functions as the operative selective pressure for a significant portion of the global population. Second, that the Darwinian reproductive directive has been functionally replaced by a manufactured achievement directive that produces behaviours directly opposed to biological fitness. Third, that the 2020 inflection point represents a measurable acceleration of Anthropogenic Evolution's biological consequences.

This is not a speculative future. It is the present. Every dietary system, every educational structure, every agricultural practice, every medical intervention, every gene editing decision, every algorithm designed to capture attention, and every cultural standard of achievement is an evolutionary act. The question is no longer whether humans are driving their own evolution. The question is whether they are doing so consciously or by accident.

No framework is complete because the thing being described has not finished happening. The selector has become the most consequential variable in its own evolution, and it does not yet know what it is selecting for.

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